

积分中值定理

设函数 $f(x), g(x)$ 在闭区间 $[a, b]$ 上连续, 且 $g(x)$ 在 $[a, b]$ 上不变号, 则存在 $\xi \in (a, b)$, 使得

$$\int_a^b f(x)g(x)dx = f(\xi) \int_a^b g(x)dx$$

不妨令 m, M 为 $f(x)$ 在 $[a, b]$ 上的 $\min$ 与 $\max$. $m \leq f(x) \leq M$, 且 $g(x) \geq 0$

$$\text{有 } m \int_a^b g(x)dx \leq \int_a^b f(x)g(x)dx \leq M \int_a^b g(x)dx$$

若 $\int_a^b g(x)dx = 0$. $\forall x \in [a, b] g(x) = 0$, 有 $\int_a^b f(x)g(x)dx = f(\xi) \int_a^b g(x)dx$

$$\text{若 } \int_a^b g(x)dx > 0 \quad m \leq \frac{\int_a^b f(x)g(x)dx}{\int_a^b g(x)dx} \leq M \quad \textcircled{1}$$

运用连续介值定理, 至少存在一点 $\xi \in [a, b]$ s.t. $f(\xi) = \frac{\int_a^b f(x)g(x)dx}{\int_a^b g(x)dx}$

下证 $\xi \in (a, b)$

若 $\textcircled{1}$ 式左右均取等或均取不等号, 结论显然成立.

若一端取等, 不妨设 $m = \frac{\int_a^b f(x)g(x)dx}{\int_a^b g(x)dx} < M$

$$\text{即 } \int_a^b (f(x) - m)g(x)dx = 0$$

因为 $f(x), g(x)$ 在 $[a, b]$ 连续. $f(x) - m \geq 0$. $g(x) \geq 0$ $(f(x) - m)g(x) = 0$

而 $\int_a^b g(x)dx > 0 \Rightarrow$ 存在开区间 $(\alpha, \beta) \subset [a, b]$ s.t. $g(x) > 0$. $f(x) - m = 0 \quad \forall x \in (\alpha, \beta)$

任取 $\xi \in (\alpha, \beta)$ 成立 $\Rightarrow$ 有 $\xi \in (a, b)$

$$\int_a^b f(x) g(x) dx = f(\xi) \int_a^b g(x) dx$$

牛顿-莱布尼茨公式

设 I 为任一闭区间, 函数 $f(x)$ 在 I 上可积, $c \in I$ 则 $\Phi(x) = \int_c^x f(t) dt$

$$\lim_{\Delta x \rightarrow 0} \frac{\Phi(x+\Delta x) - \Phi(x)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{\int_x^{x+\Delta x} f(t) dt}{\Delta x} = \lim_{\Delta x \rightarrow 0} f(\xi) \quad \begin{array}{l} \text{(积分中值定理)} \\ \text{变上限定积分} \end{array}$$

则 $\Phi(x)$ 可导, 且 $\Phi'(x) = f(x)$

$$\frac{d}{dx} \int_c^x f(t) dt = f(x)$$

$$\frac{d}{dx} \int_c^{\varphi(x)} f(t) dt = f(\varphi(x)) \cdot \varphi'(x) \quad \Leftarrow \quad \frac{d}{dx} \int_a^u f(t) dt \stackrel{\text{链式法则}}{=} \frac{d}{du} \int_a^u f(t) dt \frac{du}{dx}$$

$$\frac{d}{dx} \int_{\psi(x)}^{\varphi(x)} f(t) dt = f(\varphi(x)) \varphi'(x) - f(\psi(x)) \psi'(x)$$

$f(x)$ 在 $[a, b]$ 上连续, $F(x)$ 是 $f(x)$ 的原函数 (被积函数连续)

$$\Phi(x) = \int_a^x f(t) dt$$

$$\Phi'(x) = f(x) = F'(x) \Rightarrow \Phi(x) = F(x) + C$$

$$\int_a^b f(x) dx = \Phi(b) - \Phi(a) = F(b) - F(a)$$

例: 求 $\int_{-\pi}^{\pi} \sin mx \sin nx dx$ $\{1, \sin x, \sin 2x, \dots, \cos x, \cos 2x, \dots\}$

$$|m| \neq |n| \text{ 时. } I = \frac{1}{2} \int_{-\pi}^{\pi} (\cos(m-n)x - \cos(m+n)x) dx = \frac{1}{2} \left(\frac{\sin(m-n)x}{m-n} - \frac{\sin(m+n)x}{m+n} \right) \Big|_{-\pi}^{\pi} = 0$$

$$\text{当 } m=n \text{ 时. } I = \frac{1}{2} \int_{-\pi}^{\pi} (1 - \cos 2nx) dx = \frac{1}{2} \left(x - \frac{1}{2n} \sin 2nx \right) \Big|_{-\pi}^{\pi} = \pi$$

$$\text{当 } m=-n \text{ 时 } I = - \int_{-\pi}^{\pi} \sin^2 nx dx = -\pi$$

$$\text{当 } m=n=0 \text{ 时 } I = \int_{-\pi}^{\pi} \sin mx \sin nx dx = 0$$

例: $p>0$ 时, 证明 $\lim_{n \rightarrow \infty} \int_n^{n+p} \frac{\sin x}{x} dx = 0$

应用积分中值定理 原式 = $\sin \xi_n \ln\left(\frac{n+p}{n}\right)$, $n < \xi_n < n+p$

$$|\sin \xi_n| \leq 1, \lim_{n \rightarrow \infty} \ln\left(\frac{n+p}{n}\right) = 0$$

「典·数列化为黎曼和」

$$\text{例 } \lim_{n \rightarrow \infty} \left(\frac{1}{\sqrt{4n^2+1}} + \frac{1}{\sqrt{4n^2+4}} + \dots + \frac{1}{\sqrt{4n^2+n^2}} \right) = \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{\sqrt{4 + (\frac{k}{n})^2}} \cdot \frac{1}{n} = \int_0^1 \frac{1}{\sqrt{4+x^2}} dx$$

$$\int \frac{1}{\sqrt{4+x^2}} dx = \int \frac{2 \operatorname{ch} t}{2 \operatorname{ch} t} dt = t + C = \operatorname{arcsinh}\left(\frac{x}{2}\right) + C = \ln\left(\frac{x}{2} + \sqrt{1 + \frac{x^2}{4}}\right) + C = \ln(x + \sqrt{4+x^2}) + C$$

$$\text{原式} = \ln \frac{1+\sqrt{5}}{2}$$

定积分换元法

1. $x = \varphi(t)$ 在区间 $[\alpha, \beta]$ 或 $[\beta, \alpha]$ 上存在连续导数
2. 函数 $f(x)$ 在 $\varphi(t)$ 的值域上连续
3. $a = \varphi(\alpha)$ $b = \varphi(\beta)$

$$\int_a^b f(x) dx = \int_{\alpha}^{\beta} f(\varphi(t)) \varphi'(t) dt$$

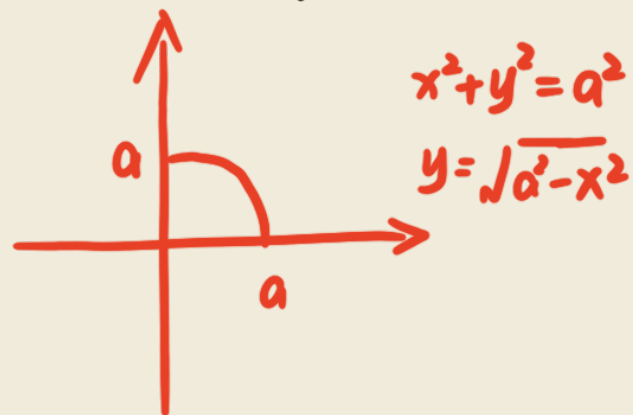
$F(x)$ 是 $f(x)$ 的原函数.

$$\int_a^b f(x) dx = F(b) - F(a)$$

$$\int_{\alpha}^{\beta} f(\varphi(t)) \varphi'(t) dt = F(\varphi(t)) \Big|_{t=\alpha}^{t=\beta} = F(\varphi(\beta)) - F(\varphi(\alpha))$$

例. $I = \int_0^a \sqrt{a^2 - x^2} dx \quad (a > 0)$

$$\underline{x = a \sin t} \quad \int_0^{\frac{\pi}{2}} a^2 \cos^2 t dt = a^2 \int_0^{\frac{\pi}{2}} \frac{1 + \cos 2t}{2} dt = \frac{a^2}{2} (t + \frac{1}{2} \sin 2t) \Big|_0^{\frac{\pi}{2}} = \frac{\pi a^2}{4}$$



设 $f(x)$ 在 $\mathbb{R}$ 上连续, 并且以 T 为周期, 证明对于任意实数 a 有 $\int_a^{a+T} f(x) dx = \int_0^T f(x) dx$

$$\int_a^{a+T} f(x) dx = \int_a^0 f dx + \int_0^T f dx + \int_T^{a+T} f dx$$

$$\text{而 } \int_T^{a+T} f(x) dx \stackrel{t=x-T}{=} \int_0^a f(t+T) dt = \int_0^a f(t) dt$$

$$\text{故 } \int_a^{a+T} f(x) dx = \int_0^T f(x) dx$$

「典·换元洛必达」

$$\lim_{x \rightarrow 0} \frac{\int_0^x \frac{\sin xt}{t} dt}{x^2} \quad (\text{对于 } x \cdot \frac{0}{0} \text{ 型洛必达}) \quad \text{直接洛必达, } x \text{ 在被积函数中}$$

$$\text{对于分子 } \begin{matrix} u=xt \\ t=\frac{u}{x} \end{matrix} \int_0^{x^2} \frac{\sin u}{\frac{u}{x}} \frac{du}{x} \quad (x \text{ 在此处为常数})$$

$$= \lim_{x \rightarrow 0} \frac{\int_0^{x^2} \frac{\sin u}{u} du}{x^2} \stackrel{\text{洛必达}}{=} \lim_{x \rightarrow 0} \frac{\frac{\sin x^2}{x^2} \cdot 2x - \frac{\sin x^2}{x^2} \cdot 3x^2}{2x^2} = 1$$

「典·循环型积分」

$$I = \int_0^\pi \frac{x \sin x}{1 + \cos^2 x} dx \stackrel{\begin{matrix} x=\pi-t \\ t=\pi-x \end{matrix}}{=} \int_\pi^0 \frac{(\pi-t) \sin t}{1 + \cos^2 t} (-dt) = \int_0^\pi \frac{(\pi-x) \sin x}{1 + \cos^2 x} dx$$

$$\Rightarrow 2I = \pi \int_0^\pi \frac{\sin x}{1 + \cos^2 x} dx \stackrel{t=\cos x}{=} \pi \int_1^{-1} \frac{-dt}{1+t^2} = \pi \arctan t \Big|_{-1}^1 = \frac{\pi^2}{2}$$

$$I = \frac{\pi^2}{4}$$

分部积分

$$I_n = \int_0^{\frac{\pi}{2}} \sin^n x dx = \int_0^{\frac{\pi}{2}} \cos^n x dx$$

$$\stackrel{t: \frac{\pi}{2}-x}{=} \int_{\frac{\pi}{2}}^0 \sin^n(\frac{\pi}{2}-t) dt = \int_0^{\frac{\pi}{2}} \cos^n x dx$$

$$I_n = - \int_0^{\frac{\pi}{2}} \sin^{n-1} x d \cos x = - \sin^{n-1} x \cos x \Big|_0^{\frac{\pi}{2}} + (n-1) \int_0^{\frac{\pi}{2}} \cos^2 x \sin^{n-2} x dx$$

$$(n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x dx \stackrel{||}{=} (n-1) \int_0^{\frac{\pi}{2}} \sin^n x dx$$

$$(n-1) I_{n-2} - (n-1) I_n$$

$$\Rightarrow I_n = \frac{n-1}{n} I_{n-2} = \frac{(n-1)(n-3)}{n \cdot (n-2)} I_{n-4} \dots$$

$$\Rightarrow I_n = \begin{cases} \frac{(n-1)!!}{n!!} \cdot \frac{\pi}{2} & n=2k \\ \frac{(n-1)!!}{n!!} & n=2k+1 \end{cases} \quad (k \in \mathbb{Z})$$

求证 $\int_0^x f(t)(x-t) dt = \int_0^x (\int_0^t f(x) dx) dt$

法一: $I_L = x \int_0^x f(t) dt - \int_0^x t f(t) dt$ $I_L' = \int_0^x f(t) dt + x \cdot f(x) - x f(x)$

$I_R = \int_0^x (\int_0^t f(u) du) dt$ $I_R' = \int_0^x f(u) du$

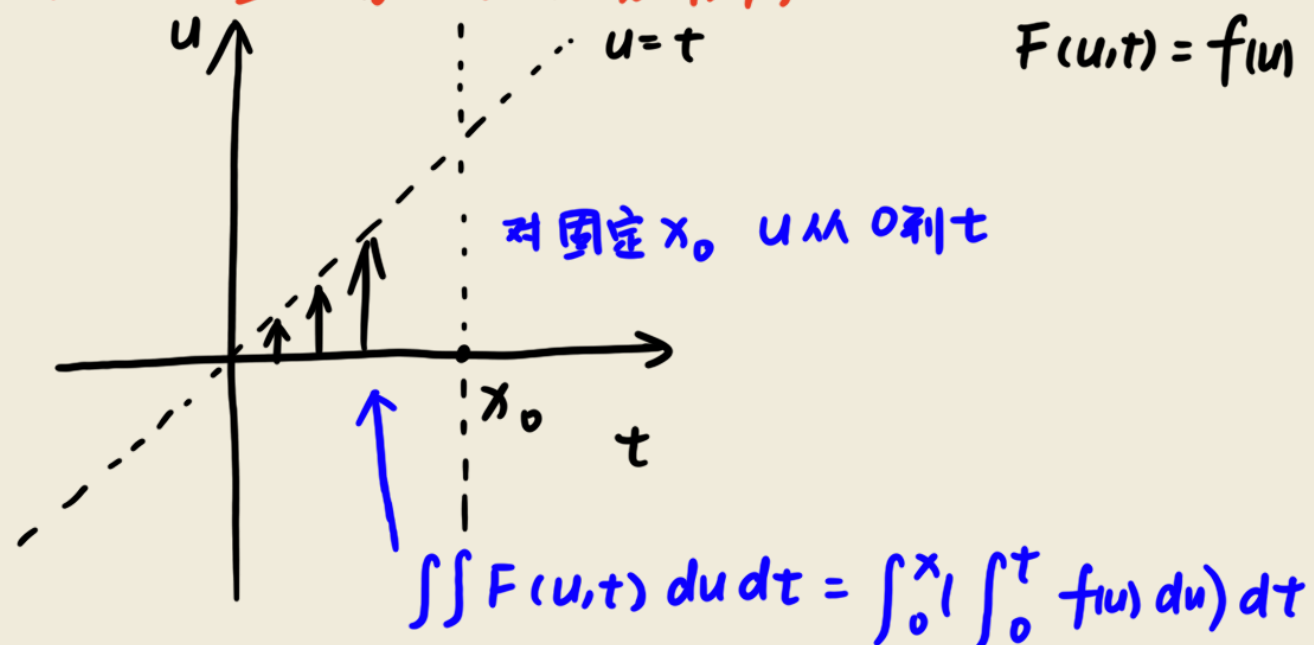
$x=0$ 时 $I_L = I_R = 0$

法二: $\int_0^x F(t) dt = \int_0^x f(t) dt$ $dF(x) = f(x) dx$

左 = $\int_0^x f(t)(x-t) dt = \int_0^x (x-t) dF(t)$

分部积分 $(x-t) \cdot F(t) \Big|_0^x - \int_0^x F(t)(-dt)$
 $= \int_0^x F(t) dt = \int_0^x \left(\int_0^t f(u) du \right) dt = \text{右}$

本质: 二重积分交换积分顺序



$$\begin{aligned} \iint F(u,t) du dt &= \int_0^{x_0} \left(\int_0^t f(u) du \right) dt \\ &= \int_0^{x_0} \left(\int_u^{x_0} f(u) dt \right) du \\ &\quad \parallel \\ &= \int_0^{x_0} f(t)(x-t) dt \end{aligned}$$